

Asignatura

Sistemas secuenciales programables

UNIDAD 1

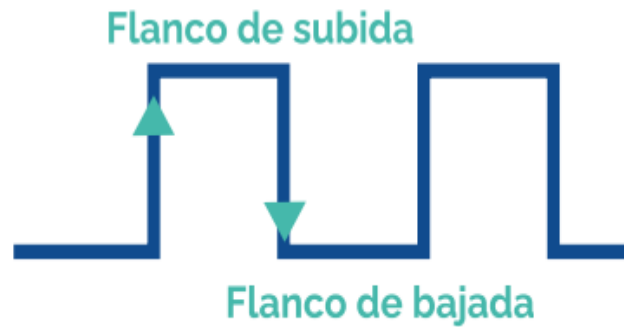
Lógica digital (I)



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Tipos de señales

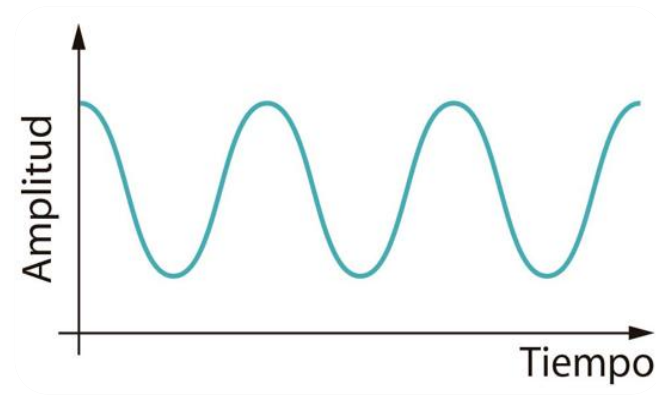
Señales digitales



Valores discretos, todo o nada,
0 o 1



Señales analógicas



Valores continuos, entre un
valor mínimo y máximo





Sistemas de numeración

Sistema decimal

Base 10

0-1-2-3-4-5-6-7-8-9

Sistema binario

Base 2

0 ó 1

Sistema octal

Base 8

0-1-2-3-4-5-6-7

Sistema hexadecimal

Base 16

Del 0 al 9
De la A a la F

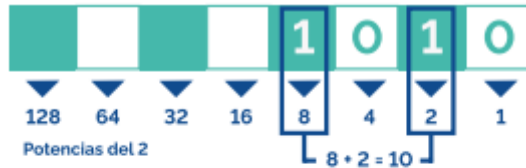
Equivalencias entre sistemas

Decimal	Binario	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F



Conversiones de binario a decimal y viceversa

Binario a decimal



Binario decimal
11010011 = 211

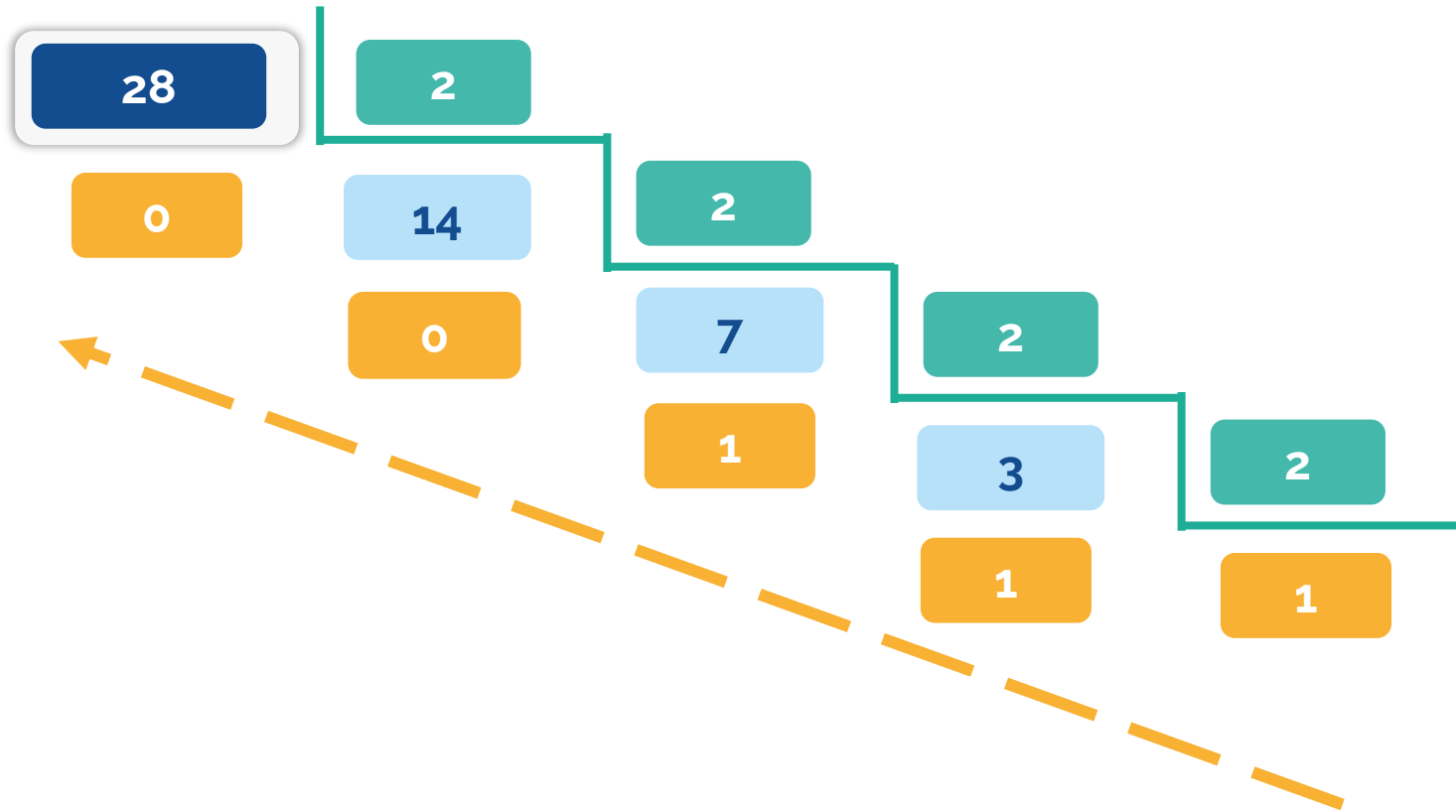
Decimal a binario



37 = 100101

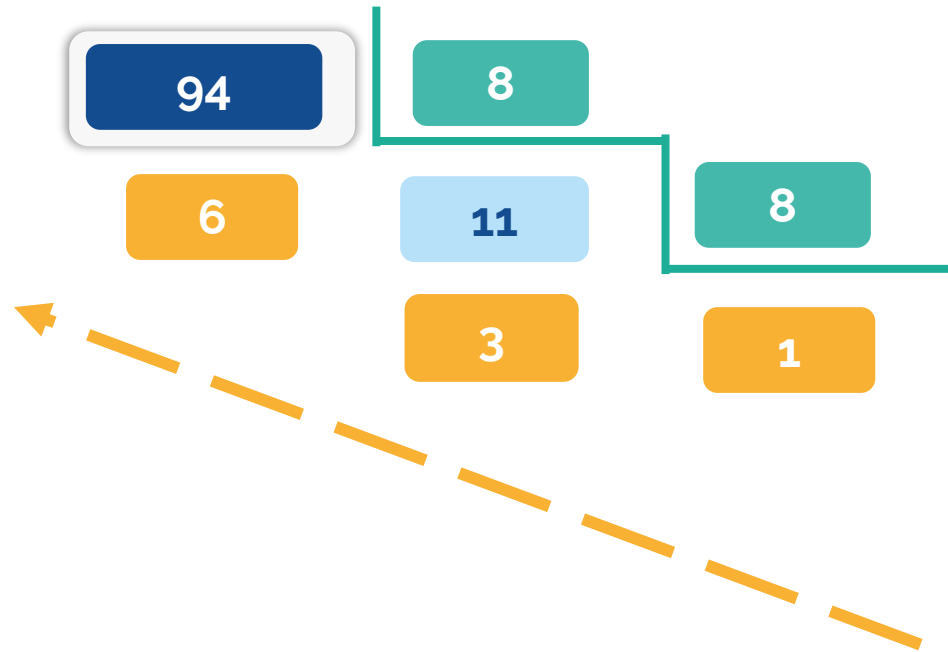
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8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Pasar a binario el número 28 en base decimal



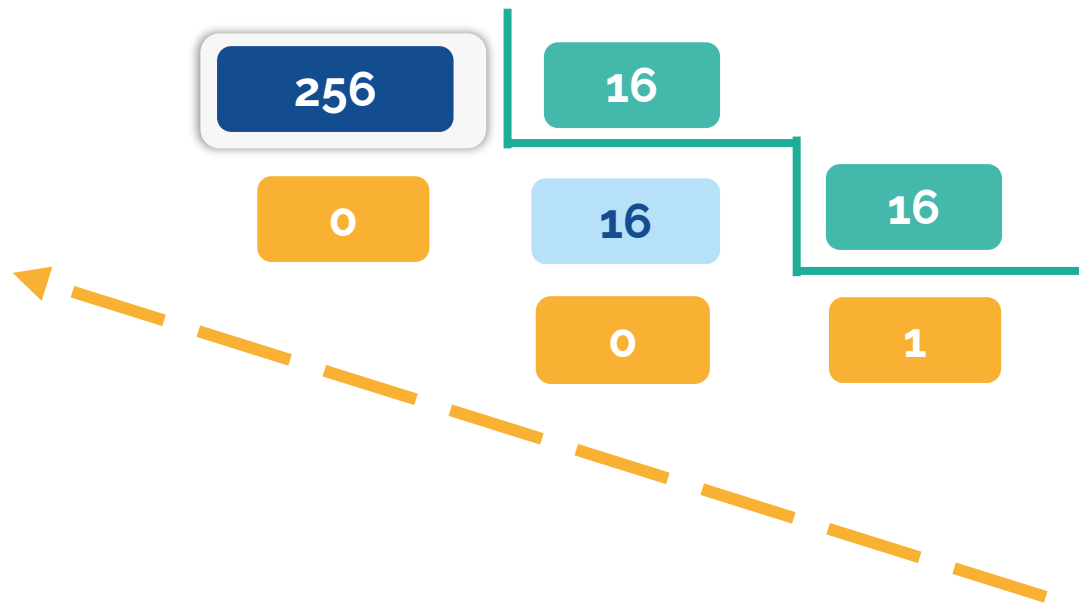
$$28_{(10)} = 11100_{(2)}$$

Pasar a octal el número 94 en base decimal



$$\begin{array}{|c|} \hline 94 \\ \hline \end{array} \begin{array}{|c|} \hline (10) \\ \hline \end{array} = \begin{array}{|c|} \hline 1 \\ \hline \end{array} \begin{array}{|c|} \hline 3 \\ \hline \end{array} \begin{array}{|c|} \hline 6 \\ \hline \end{array} \text{ Base } \begin{array}{|c|} \hline (8) \\ \hline \end{array}$$

Pasar a hexadecimal el número 256 en base decimal



$$256_{(10)} = 100_{(16)}$$

Base (16)

Pasar el número 110110 en binario a base 10, 8 y 16



A decimal

1 1 0 1 1 0

$\downarrow \times$ $\downarrow \times$ $\downarrow \times$ $\downarrow \times$ $\downarrow \times$ $\downarrow \times$
 2^5 2^4 2^3 2^2 2^1 2^0

$$(2^0 \times 0) = 1 \times 0 = 0$$

$$(2^1 \times 1) = 2 \times 1 = 2$$

$$(2^2 \times 1) = 4 \times 1 = 4$$

$$(2^3 \times 0) = 8 \times 0 = 0$$

$$(2^4 \times 1) = 16 \times 1 = 16$$

$$(2^5 \times 1) = 32 \times 1 = 32$$

$\downarrow +$
 $0 + 2 + 4 + 0 + 16 + 32 =$ **54**

A octal

1 1 0 1 1 0

$\downarrow \times$ $\downarrow \times$ $\downarrow \times$
 2^2 2^1 2^0

$$(2^0 \times 0) = 1 \times 0 = 0$$

$$(2^1 \times 1) = 2 \times 1 = 2$$

$$(2^2 \times 1) = 4 \times 1 = 4$$

6

$6_{(10)}$

$6_{(10)}$

8

6

0

$6_{(8)}$

$6_{(8)}$

66₍₈₎

A hexadecimal

1 1 0 1 1 0

$3_{(10)}$ **16** $6_{(10)}$ **16**
3 **0** **6** **0**

$\downarrow$

$\downarrow$

$3_{(16)}$

$6_{(16)}$

36₍₁₆₎



568

0101 0110 1000

El código BCD estará compuesto por 4 bits binarios y es el que más usaremos en digital

[illegible]

The background is a solid blue color. Overlaid on this are several faint, light-blue geometric patterns. These include a grid of small squares that form larger, irregular shapes, and numerous small, light-blue arrows pointing in various directions. The overall effect is a sense of movement and digital connectivity.

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